

MongoDB Multi-Collection Assignment

```
## Database
```

```
```
```

```
retail_db
```

```
```
```

```
Collections:
```

```
    * `customers`
```

```
    * `products`
```

```
    * `orders`
```

```
---
```

```
# Step 1 — Create Customers Collection
```

```
```javascript
```

```
db.customers.insertMany([
```

```
{ customer_id:1, name:"Arjun", city:"Hyderabad" },
```

```
{ customer_id:2, name:"Rahul", city:"Bangalore" },
```

```
{ customer_id:3, name:"Sneha", city:"Mumbai" },
```

```
{ customer_id:4, name:"Amit", city:"Delhi" },
```

```
{ customer_id:5, name:"Priya", city:"Hyderabad" }
```

```
])
```

```
Step 2 — Create Products Collection
```

```
```javascript
```

```
db.products.insertMany([
```

```
{ product_id:101, name:"Laptop", category:"Electronics", price:75000 },
```

```
{ product_id:102, name:"Phone", category:"Electronics", price:50000 },
```

```
{ product_id:103, name:"Desk", category:"Furniture", price:15000 },
```

```
{ product_id:104, name:"Chair", category:"Furniture", price:7000 },
```

```
{ product_id:105, name:"Tablet", category:"Electronics", price:30000 }
```

```
])
```

```
```
```

### # Step 3 — Create Orders Collection

```
```\javascript
db.orders.insertMany([
  { order_id:1001, customer_id:1, product_id:101, quantity:1, order_date:"2024-03-01" },
  { order_id:1002, customer_id:2, product_id:102, quantity:1, order_date:"2024-03-02" },
  { order_id:1003, customer_id:1, product_id:105, quantity:2, order_date:"2024-03-03" },
  { order_id:1004, customer_id:3, product_id:103, quantity:1, order_date:"2024-03-05" },
  { order_id:1005, customer_id:5, product_id:102, quantity:3, order_date:"2024-03-07" }
])
```
```

## MongoDB Assignment Questions

### Basic Queries

1. Display all customers.

**db.customers.find()**

2. Display all products.

**db.products.find()**

3. Display all orders.

**db.orders.find()**

---

### Filtering

4. Show all customers from **\*\*Hyderabad\*\***.

**db.customers.find({city: "Hyderabad"});**

5. Show all **\*\*Electronics products\*\***.

**db.products.find({category: "Electronics"});**

6. Show products whose **\*\*price is greater than 30000\*\***

**db.products.find({price: {\$gt: 30000}});**

.

7. Show orders where **\*\*quantity is greater than 1\*\***.

**db.orders.find({quantity: {\$gt: 1}});**

---

## Sorting

8. Display products **\*\*sorted by price descending\*\***.

```
db.products.find().sort({price:-1});
```

9. Display customers **\*\*sorted alphabetically\*\***.

```
db.customers.find().sort({name:1});
```

---

## Aggregation

10. Count the **\*\*total number of orders\*\***.

```
db.orders.aggregate([
 {
 $group: {
 _id: null,
 totalOrders: {$sum:"$quantity"}
 }
 }
]);
```

11. Find the **\*\*average price of products\*\***.

```
db.products.aggregate([
 {
 $group: {
 _id: null,
 avg_Price: {$avg:"$price"}
 }
 }
]);
```

12. Find the **\*\*maximum product price\*\***.

```
db.products.aggregate([
 {
 $group: {
 _id: null,
 maxi_Price: {$max: "$price"}
 }
 }
])
```

13. Find the **\*\*total quantity ordered per product\*\***.

```
db.orders.aggregate([
 {
 $group: {
 _id: "$product_id",
 totalQuantity: {$sum: "$quantity"}
 }
 }
]);
```

---

## Multi-Collection Queries (\$lookup)

14. Display \*\*orders along with customer details\*\*.

```
db.orders.aggregate([
 {
 $lookup: {
 from: "customers",
 localField: "customer_id",
 foreignField: "customer_id",
 as: "customer_details"
 }
 }
])
```

15. Display \*\*orders along with product details\*\*.

```
db.orders.aggregate([
 {
 $lookup: {
 from: "products",
 localField: "product_id",
 foreignField: "product_id",
 as: "product_details"
 }
 }
])
```

16. Display **\*\*customer name and products they ordered\*\***.

```
db.orders.aggregate([
 {
 $lookup: {
 from: "customers",
 localField: "customer_id",
 foreignField: "customer_id",
 as: "cust"
 }
 },
 {
 $lookup: {
 from: "products",
 localField: "product_id",
 foreignField: "product_id",
 as: "prod"
 }
 },
 { $unwind: "$cust" },
 { $unwind: "$prod" },
 {
 $project: {
 _id: 0,
 customerName: "$cust.name",
 productName: "$prod.name"
 }
 }
])
```

17. Display **\*\*product name and total quantity sold\*\***.

```
db.orders.aggregate([
 {
 $group: {
 _id: "$product_id",
 totalQuantitySold: { $sum: "$quantity" }
 }
 },
 {
 $lookup: {
 from: "products",
 localField: "_id",
 foreignField: "product_id",
 as: "product_info"
 }
 },
 {
 $unwind: "$product_info"
 },
 {
 $project: {
 _id: 0,
 productName: "$product_info.name",
 totalQuantitySold: 1
 }
 }
])
```

---



## Advanced Aggregation

18. Calculate **\*\*total revenue per product\*\***.

```
db.orders.aggregate([
 {
 $lookup: {
 from: "products",
 localField: "product_id",
 foreignField: "product_id",
 as: "prod"
 }
 },
 { $unwind: "$prod" },
 {
 $group: {
 _id: "$prod.name",
 totalRevenue: { $sum: { $multiply: ["$quantity", "$prod.price"] } }
 }
 }
])
```

19. Calculate **\*\*total revenue generated by each customer\*\***.

```
db.orders.aggregate([
 {
 $lookup: {
 from: "products",
 localField: "product_id",
 foreignField: "product_id",
 as: "prod"
 }
 }
```

```
},
{ $unwind: "$prod" },
{
 $group: {
 _id: "$customer_id",
 revenue: { $sum: { $multiply: ["$quantity", "$prod.price"] } }
 }
},
{
 $lookup: {
 from: "customers",
 localField: "_id",
 foreignField: "customer_id",
 as: "cust"
 }
},
{ $unwind: "$cust" },
{
 $project: {
 _id: 0,
 customerName: "$cust.name",
 totalSpent: "$revenue"
 }
}
}
```

D)

20. Find the **\*\*most sold product\*\***.

```
db.orders.aggregate([
 {
 $group: {
 _id: "$product_id",
 totalSold: { $sum: "$quantity" }
 }
 },
 { $sort: { totalSold: -1 } },
 { $limit: 1 },
 {
 $lookup: {
 from: "products",
 localField: "_id",
 foreignField: "product_id",
 as: "info"
 }
 },
 { $unwind: "$info" },
 {
 $project: {
 _id: 0,
 productName: "$info.name",
 totalSold: 1
 }
 }
])

```